

HaDy_MZB

Habitat Dynamics MacroZooBenthos

A Python toolbox for patch-scale habitat dynamics analysis

User manual

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GitHub repository:

https://github.com/HaDy-toolbox/HaDy_MZB

DOI: XXXX

Impressum

Authors of the toolbox and user manual

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Scientific background

This toolbox implements the framework presented in: Lecrivain A, De Cesare G, Weber C, & Bätz N (in Review). Macroinvertebrate habitat dynamics under frequent hydropower-induced discharge fluctuations: patch-scale metrics to quantify effects of hydropeaking and morphological complexity. XXXX

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XXX

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1. Toolbox presentation

1.1 Purpose of the HaDy_MZB toolbox

Rivers are dynamic and habitat conditions change over time as discharge varies and interacts with channel morphology. These changes are especially relevant at the patch scale, where organisms experience hydraulic stress, habitat persistence, dewatering, and shifts between suitable and unsuitable habitat conditions.

HaDy_MZB is a free and open source Python-based [HaDy-toolbox](#) for quantifying flow-driven habitat dynamics at the patch scale, with a focus on benthic macroinvertebrates—organisms particularly vulnerable due to their limited mobility, complex life cycles, and sensitivity to hydraulic stress. However, the toolbox can also be adapted to other organisms such as fish eggs, larvae, or vegetation, which may likewise be affected by limited mobility or exposure to changing habitat conditions.

HaDy_MZB links flow time-series with precomputed hydraulic or habitat information for selected discharges to reconstruct habitat time-series. Based on these habitat time-series, the toolbox calculates spatially explicit metrics that describe habitat availability, habitat stability, and exposure to hydraulic stress (see Table 1). A patch represents a fixed spatial unit corresponding to the ecological perception of macroinvertebrates, typically defined by the computational mesh used in high-resolution hydrodynamic models.

The toolbox builds on the patch-scale approach developed by Bätz and applied in Lecrivain. In particular, it puts into practice the logic of deriving habitat time-series from flow scenarios and habitat distributions. Hydropeaking is a key application because frequent sub-daily flow fluctuations strongly increase habitat dynamics, but the toolbox can also be used more generally to analyze habitat responses to variable flow regimes.

Table 1 Metrics derived by the HaDy_MZB toolbox

Metric	Name	What it measures	Ecological meaning
M1	Habitat probability	Time-integrated availability of habitat conditions	Indicates habitat availability and dominance
M2	Habitat shifts	Frequency of shifts between habitats	Reflects habitat stability and persistency
M3	Drift risk	Exposure to hydraulically stressful conditions (e.g. high current velocity)	Indicates risk of drift
M4	Desiccation risk	Exposure to low water depth / drying conditions	Indicates risk of stranding/dessciation

Who HaDy_MZB is for:

HaDy_MZB is designed for **researchers** and **practitioners** who want to assess flow-driven habitat dynamics and ecological stressors in a spatially explicit way. It supports applications such as hydropeaking impact assessment, comparison of hydrological scenarios, evaluation of morphological complexity, and reach-scale mitigation planning.

The toolbox does not replace hydrodynamic modelling or ecological expert judgement. Instead, it provides a reproducible workflow for translating existing hydraulic or habitat data and flow time-series into interpretable habitat-dynamics metrics.

1.2 Overview of the Workflow

The execution of the toolbox involves 3 steps (Figure 1):

- **Pre-processing:** required input data (flow time-series, hydraulic or habitat data) are collected and formatted to prepare the analysis (see section 3). Based on these inputs, different scenarios can be configured and compared.
- **Processing:** the toolbox links flow time-series with hydraulic or habitat data to generate habitat time-series and calculate selected metrics.
- **Post-processing:** the resulting outputs can be visualized, mapped and further analysed using GIS software such as QGIS (see section 5 , and 9 QGIS tips and tricks).

A project is defined for a given study area where hydrodynamic simulations have already been carried out by the user. From the simulations, spatially distributed data on water depth and current velocity must be provided by the user for a representative range of discharges (input shapefile). These data are then used to derive habitat types. Alternatively, the user can directly provide a spatially explicit habitat classification for a representative range of discharges. In addition, a flow time-series is required to describe the temporal evolution of discharge and define the simulation timestep.

HaDy_MZB does not run hydrodynamic simulations. Instead, it uses existing hydraulic or habitat data and user-defined ecological criteria to quantify habitat dynamics in a reproducible way. The respective roles of the user and the toolbox are summarized in Table 2.

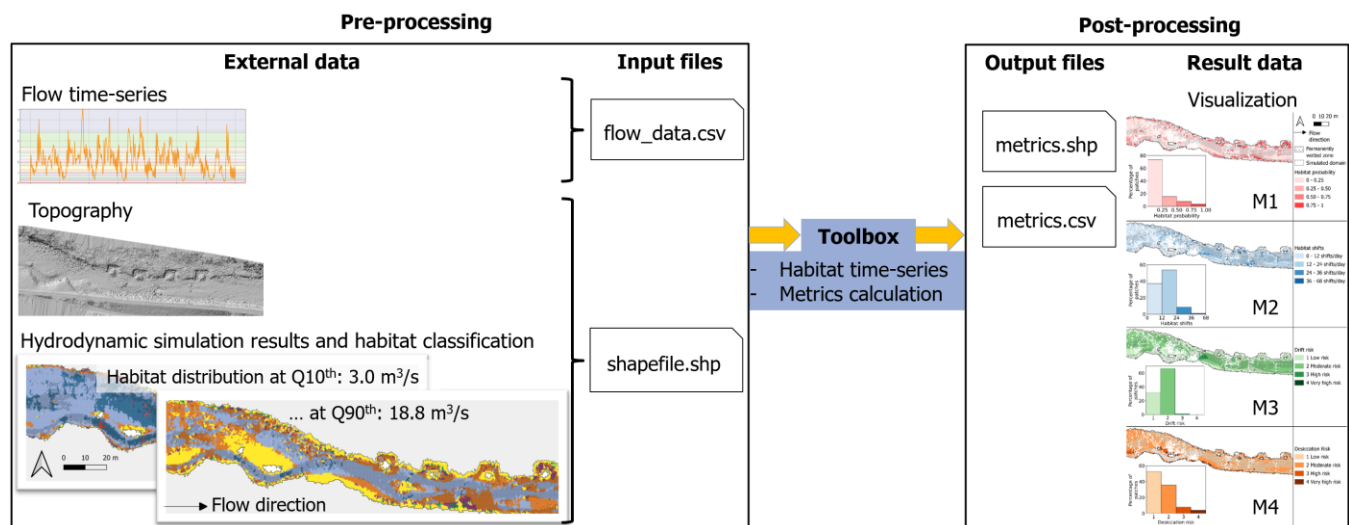


Figure 1 Overview of the execution procedure of the toolbox HaDy_MZB

Table 2 Respective roles of the user and the toolbox throughout the workflow

User responsibilities	
Pre-processing	Selects the flow time-series at the timestep of interest
	Selects associated discharges of interest
	Selects the zone of interest
	Runs the hydrodynamic simulation for the zone and representative discharges
	Decides on a habitat classification method and relevant ecological thresholds
Post processing	Analyses the output on GIS software
	Adds additional plots or analysis if needed
Toolbox responsibilities	
Processing	Builds habitat time-series from flow time-series, water depth, current velocity and related habitat classification
	Calculates the selected metrics for the habitat type of interest
	Exports results (i.e. metric values) in a shapefile and a csv document
	Provides graphical plots
	Provides additional functions for deeper analysis if needed

1.3 Example case study

This manual follows a running example dataset provided with the toolbox (Windows operating system). Blue boxes such as the following provide practical examples and shortcuts to quickly explore, reproduce, and adapt the workflow.

Running the example case as an initial analysis is a good way to become familiar with the toolbox.

Example case study: Quick Input–Output example (no setup required)

Example inputs (data > input, see Figure 2):

- Flow time-series (see Week_timestep_10_min.csv)
- Input shapefile with water depth and current velocity value (see Example_input_shapefile.shp → column v_discharge and d_discharge)
- Habitat definition (see section 4.3 Classify habitats of this user manual: classification entered in habitat_classification.py in the scripts)

Example outputs (data > output > Example_output, see Figure 2):

- Metrics M1–M4 (see metrics.csv)
- Maps (see metrics.shp and GIS_project)
- Plots (see folder Plots)

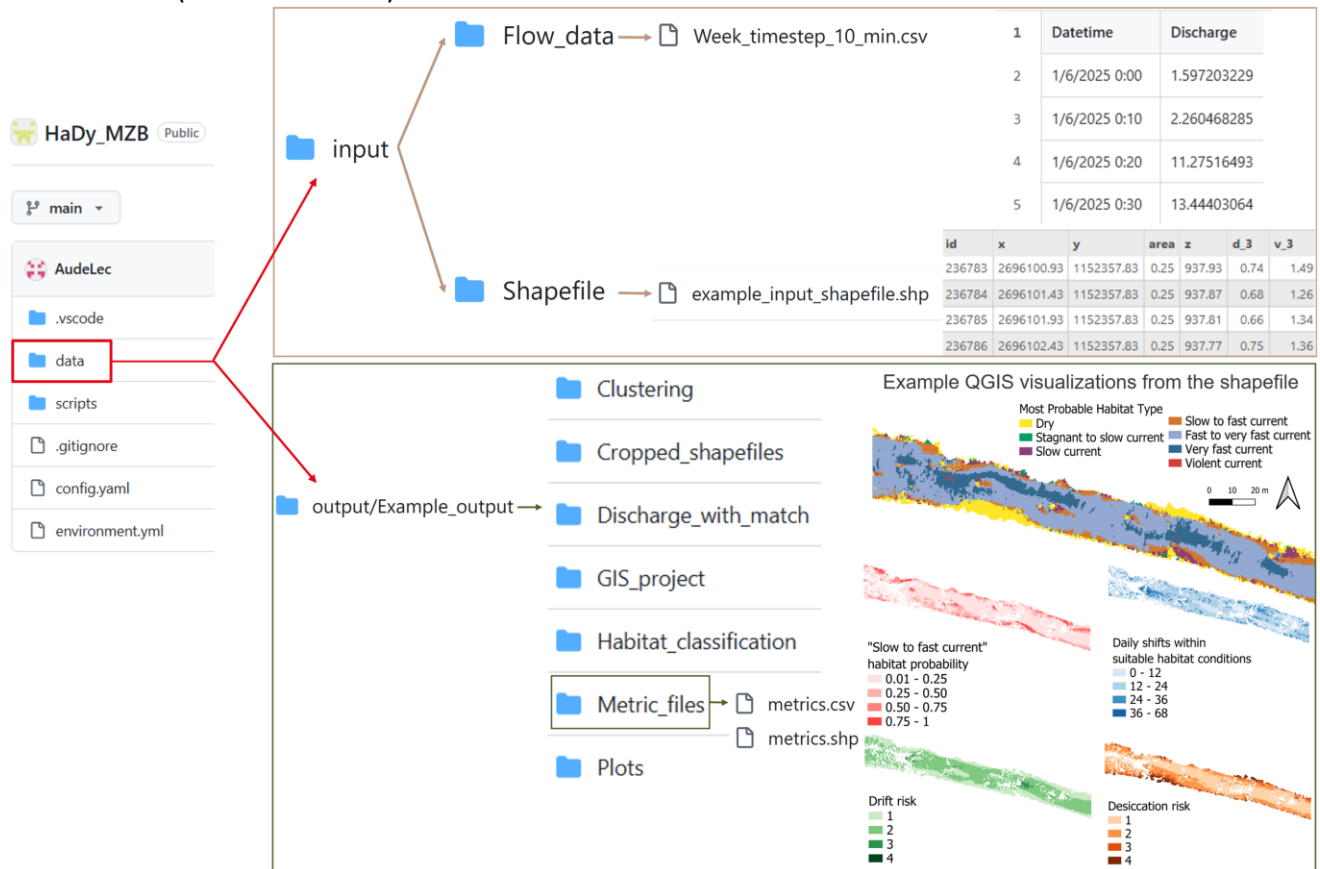


Figure 2 Quick input-output overview of the example case study GitHub

Click on the boxes Figure 3 to quickly get started with the toolbox, following the example case study blue boxes.

Quick start: key steps

2. Technical setup

3. Inputs + configuration

5. Output interpretation

Figure 3 Key steps to get started with the toolbox

2. Technical setup

This chapter explains how to install the toolbox and verify that everything works by reproducing the example dataset provided in the repository.

2.1 Prerequisites

If you need support in installing the usual python core tools mentioned below, **check the Appendix technical support file** provided in the repository.

Required tools:

- Python
- Git
- Anaconda/Miniconda
- Visual Studio Code (recommended)
- Python extension for VS Code

This file also provides support for basic commands such as accessing a folder via the terminal.

2.2 GitHub repository setup

All steps are explained and illustrated for the example case study in the blue box below.

Step 1: Create a working directory

Create a folder where the Git repository will be cloned.

This directory can be located anywhere on your system and named according to your preference.

Step 2: Clone the repository

Open a terminal and go to the created working directory: enter the command “cd” and the absolute path to your directory (**cd path/to/your/directory**).

Once in the working directory, paste the cloning command (Git needs to be installed otherwise this will not work, see 2.1 Prerequisites):

```
git clone https://github.com/HaDy-toolbox/HaDy_MZB.git
```

Step 3: Setup the virtual environment

In the terminal (for macOS and Linux), or anaconda prompt (Windows) go to **the cloned folder called HaDy_MZB** in your working directory, and paste following commands:

```
conda env create -f environment.yml, and then conda activate Venv_HaDy_MZB
```

Step 4: Open the project on VS code and select the python interpreter

Open the software VS code previously installed, open the cloned HaDy_MZB folder.

To select the interpreter: Ctrl+Shift+P (to get the command panel) → type “Python: Select Interpreter” → Choose Venv_HaDy_MZB.

2.3 Running the code

To run the main code, go the python file *main.py* and click the play button (see example box). The same steps can be done in the *plot.py* or *clustering.py* for further analysis (see section 4.6 Additional functions).



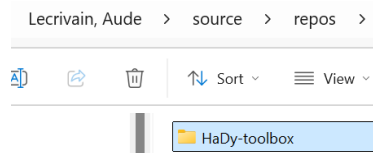
If you tailor the analysis, make sure to have adapted **and saved** your local.yaml file (more in section 3.2), otherwise outputs will not match the entered parameters.

2.4 Example case study: from python repository to running the code

The following box explains and details the steps to follow to run the example case study and get familiar with the toolbox.

Example case study: From python repository setup to running the code

Step 1: Creation of a working directory



Step 2: Cloning the repository

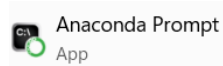
In the terminal, type in “cd” and paste the absolute path to your directory

```
PS C:\Users\lecrivau> cd C:\Users\lecrivau\source\repos\HaDy-toolbox
```

Then type in the cloning command

```
PS C:\Users\lecrivau\source\repos\HaDy-toolbox> git clone https://github.com/HaDy-toolbox/HaDy_MZB.git
```

Step 3: Setting up the virtual environment



On windows, use Anaconda prompt

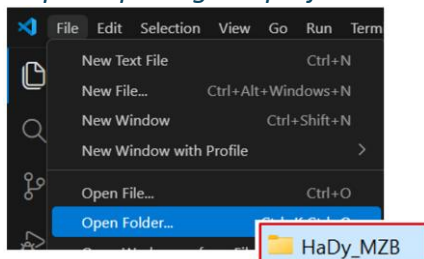
Go to the cloned HaDy_MZB within your directory and type in the commands

```
C:\Users\lecrivau\source\repos\HaDy-toolbox>cd C:\Users\lecrivau\source\repos\HaDy-toolbox\HaDy_MZB
C:\Users\lecrivau\source\repos\HaDy-toolbox\HaDy_MZB>conda env create -f environment.yml
C:\Users\lecrivau\source\repos\HaDy-toolbox\HaDy_MZB>conda env create -f environment.yml
```

You now see that the environment is activated.

```
(Venv_HaDy_MZB) C:\Users\lecrivau>
```

Step 4: Opening the project on VS code and selecting the python interpreter

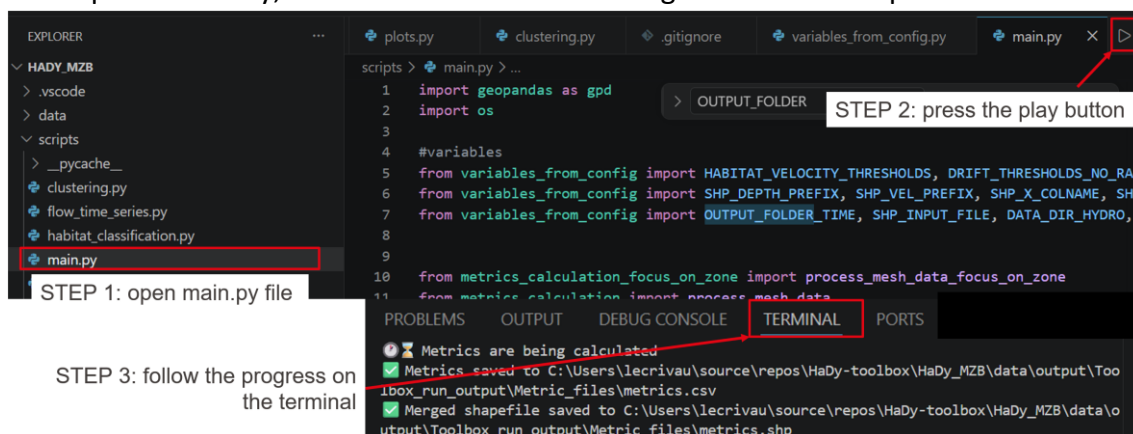


To select the interpreter: Ctrl+Shift+P (to get the command panel) → type “Python: Select Interpreter” → Choose

```
Python 3.11.11 (Venv_HaDy_MZB) ~\conda\envs\Venv_HaDy_MZB\python.exe
```

Running the code

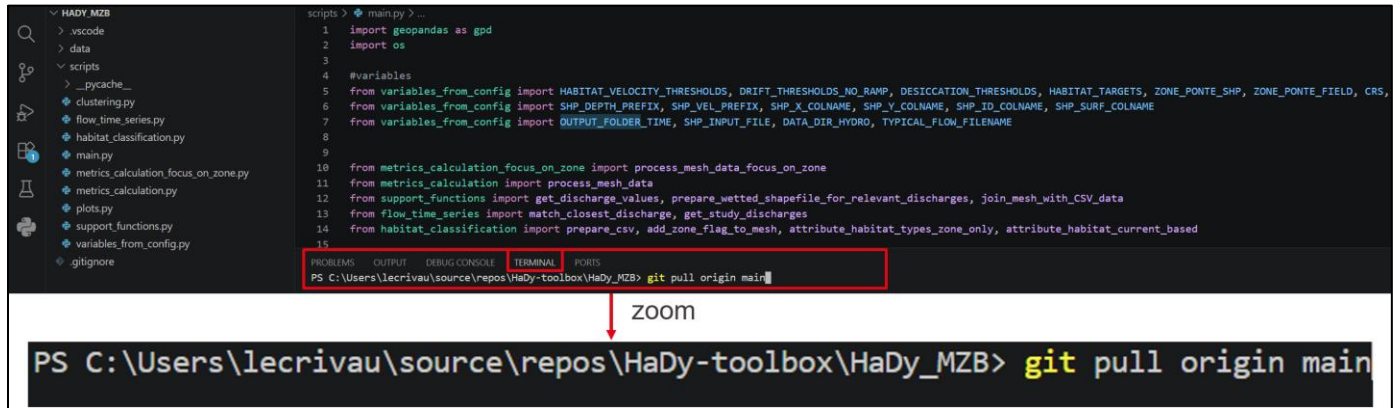
For the example case study, not modification of the configuration file is required



2.5 Later step: getting the latest version from the GitHub

Since the toolbox is in constant evolution, make sure to use the latest available. To update it, you need to “pull” the changes from the GitHub folder as done on Figure 4.

This can be done directly in the Visual Studio code terminal as follows: command **git pull origin main**



The screenshot shows the Visual Studio Code interface. On the left is the Explorer sidebar with a file tree for a project named 'HaDY_MZB'. The main editor area displays a Python script with various imports and function calls. At the bottom, the 'TERMINAL' tab is active, showing a command prompt with the command `git pull origin main` entered. A red box highlights the terminal area, and a red arrow points from it to a zoomed-in view of the same command prompt below.

```
scripts > main.py > ...
1 import geopandas as gpd
2 import os
3
4 #variables
5 from variables_from_config import HABITAT_VELOCITY_THRESHOLDS, DRIFT_THRESHOLDS_NO_RAMP, DESICCATION_THRESHOLDS, HABITAT_TARGETS, ZONE_PONTE_SHP, ZONE_PONTE_FIELD, CRS,
6 from variables_from_config import SHP_DEPTH_PREFIX, SHP_VEL_PREFIX, SHP_X_COLNAME, SHP_Y_COLNAME, SHP_ID_COLNAME, SHP_SURF_COLNAME
7 from variables_from_config import OUTPUT_FOLDER_TIME, SHP_INPUT_FILE, DATA_DIR_HYDRO, TYPICAL_FLOW_FILENAME
8
9
10 from metrics_calculation_focus_on_zone import process_mesh_data_focus_on_zone
11 from metrics_calculation import process_mesh_data
12 from support_functions import get_discharge_values, prepare_wetted_shapefile_for_relevant_discharges, join_mesh_with_csv_data
13 from flow_time_series import match_closest_discharge, get_study_discharges
14 from habitat_classification import prepare_csv, add_zone_flag_to_mesh, attribute_habitat_types_zone_only, attribute_habitat_current_based
15
```

zoom

```
PS C:\Users\lecrivau\source\repos\HaDY-toolbox\HaDY_MZB> git pull origin main
```

Figure 4 Example of how to pull the last version



ATTENTION

This command will overwrite your script folder and input data local folder with the latest content of the online GitHub repository, so **make sure to have saved ALL your changes in a safe space (e.g., a separate folder) before running this command.**

3. Inputs and configuration

3.1 Inputs files

Flow time-series input data

Necessary minimum requirements:

- csv file with variations of discharge (m^3/s) in time (e.g., Figure 5).
- The location is folder data > input > Flow_data.
- The column containing the discharges need to be called “Discharge” (exactly like that, including the capital letter). The format of the other columns is not important, and can be kept or deleted (e.g., Figure 5, right).
- Discharge values must be formatted as numbers, with decimals separated by a dot “.”
- The timestep of the discharge data entered in the configuration file must match this file’s timestep (more in section 3.2). **It is the user’s responsibility to provide an input file matching the relevant analysis timestep.**



Common issues:

- Wrong location of the file
- File as excel instead of csv format
- Wrong “Discharge” column name
- Coma instead of point as decimal separator
- Missing values
- Unmatching timesteps leading to false metric values

Date (TU)	Valeur (en m^3/s)	Statut	Qualification	MÃ©thode	ContinuitÃ©		Date (TU)	Discharge
2025-06-24T00:00:00.000Z	14.4	12	20	8	0		2025-06-24T00:00:00.000Z	14.4
2025-06-24T00:10:00.000Z	14.4	12	20	8	0		2025-06-24T00:10:00.000Z	14.4
2025-06-24T00:20:00.000Z	14.4	12	20	8	0		2025-06-24T00:20:00.000Z	14.4
2025-06-24T00:30:00.000Z	14.4	12	20	8	0		2025-06-24T00:30:00.000Z	14.4

Figure 5 Example of data downloaded from the HydroPortail platform (on the left), and what is kept and renamed for the analysis (right)

Shapefile input data

Necessary minimum requirements:

- Polygon shapefile (.shp and associated other file extensions) with water depth and current velocity data for discharges of interest (e.g., Figure 8). This data generally comes from hydrodynamic simulations
- The location is folder data > input > Shapefile
- The exact column names prefixes must be entered in the configuration file (more in section 3.2).
- The shapefile needs to contain the data presented in Table 3.

Table 3 Data that needs to be in the input polygon shapefile

Id	Id of the mesh - One row per patch
x	Coordinates of the patch center
y	
Area m2	If patches have different areas, and the user want to normalize metrics to the area
Water depth for each discharge (in meters)	Depth values for each discharge of interest. Decimals separated by “_” e.g., d_31_8 and d_6 for respectively discharges 31.8 and 6 m ³ /s if the prefix is “d_”
Water current velocity for each discharge (in meters/second)	Current velocity values for each discharge of interest. Decimals separated by “_” e.g., v_31_8 and v_6 for respectively discharges 31.8 and 6 m ³ /s if the prefix is “v_”

id	x	y	area	z	d_1	v_1	d_2	v_2	d_3	v_3	d_4_2	v_4_2	d_5_5	v_5_5	d_7_4	v_7_4	d_9	v_9	d_10_6	v_10_6	d_12	v_12
276139	2696154.44	1152347.31	0.25	937.3	0.33	0.62	0.43	0.88	0.53	1.18	0.58	1.29	0.65	1.43	0.74	1.57	0.8	1.65	0.87	1.71	0.92	1.76
276140	2696154.94	1152347.31	0.25	937.3	0.34	0.6	0.45	0.87	0.55	1.17	0.59	1.28	0.66	1.42	0.75	1.56	0.82	1.65	0.88	1.71	0.93	1.76
276141	2696155.44	1152347.31	0.25	937.28	0.35	0.57	0.46	0.82	0.56	1.11	0.61	1.21	0.68	1.36	0.77	1.5	0.84	1.59	0.9	1.65	0.95	1.7

Figure 6 Example of data in the polygon shapefile



ATTENTION

Common issues:

- Missing files: if only the .shp file in the data > input > Shapefile, the toolbox will not proceed. The other necessary formats are required (see files in data > input > Shapefile > Example_input_shapefile)
- Empty or n/a values: to be set to 0 for the toolbox to run an analysis.
- Data consistency: for a mesh having no water for a certain discharge (depth = 0), then the current velocity value also needs to be zero.
- Missing data: each discharge of interest needs to have current velocity and water depth data (metrics need both to be calculated, so just current velocity or just water depth will not be sufficient).
- Wong formatting: the water depth and current velocity fields need to have the value of the discharge of interest in their title. Floats need to be separated by a “_” (e.g., d_31_8 and v_31_8 or d_6 and v_6). The water depth and current velocity columns prefixes (before the discharge value) need to be identical between each water depth column, and between each current velocity column, and match what is entered in the configuration file (more in section 3.2).
- Continuous spatial representation of the results: it is necessary for the shapefile to be a polygon shapefile and not a point shapefile.

3.2 Configuration

Necessary minimum requirements:

- To run an analysis that is different than the example case study, a *local.yaml* needs to be created in the repository. It is the user's responsibility to create this file by copy pasting the *config.yaml* and making the corresponding modifications to tailor the analysis (see relevant video).
- The location of *local.yaml* is the root of the repository, where *config.yaml* is located



Common issues:

- Wrong name: It is the user's responsibility to create the *local.yaml* (spelled exactly that way) file to tailor the parameters.
- No *local.yaml* file: If no *local.yaml* file exists, the toolbox is running the *config.yaml* which is the example case study by default.
- Analysis not matching the entered parameters: It is important to **save the file** once changes are made to run the code with the correct parameters.
- Losing the previous analysis: Each new toolbox run stores the outputs in data > outputs > Toolbox_run_output, overwriting what is already in the folder. Save the analysis files you want to keep in another folder or rename them to keep them.

Table 4Habitat classification thresholds describes the variables that need to be adapted by the user. It is split in different categories.

Table 4 Detail of the configuration variables

Input files

Name of the variable	Signification	Status
input_shp_filename	Name of the input shapefile located in the data > input > Shapefile folder	Mandatory
crs	CSR of the GIS project	Mandatory
input_flow_timeseries_filename	Name of the flow time-series input file located in the data > input > Flow_data folder	Mandatory
time_step_min	Timestep of the flow time-series in minutes (float, e.g., 10.0 for a 10-minute timestep). ./!\ Must be the one of the flow time-series file	Mandatory
shp_id_colname	Name of the id column identifying each mesh	Mandatory
shp_x_colname	Name the x coordinate column of each mesh in the input shapefile	Mandatory
shp_y_colname	Name the y coordinate column of each mesh in the input shapefile	Mandatory

shp_area_colname	Name of the column containing the area of each mesh (for potential area metrics normalization)	Mandatory
shp_depth_prefix	Depth prefix before the discharge values in the input shapefile (e.g., "d_30_2" → the prefix is "d_")	Mandatory
shp_vel_prefix	Current velocity prefix before the discharge values in the input shapefile (e.g., "v_30_2" → the prefix is "v_")	Mandatory

Habitat classification thresholds

Name of the variable	Signification	Status
depth_threshold	Minimum water depth (in meters) to consider a mesh wet (float, e.g., 0.01 for 1 cm, meaning that meshes will be considered wet if water depth is greater than 1 cm)	Mandatory
habitat_velocity_thresholds	Example of habitat classification thresholds based on current velocity and water depth (see section 4.3 Classify habitats)	Mandatory
number_of_habitats	Integer: the number of habitats that you have based on the habitat classification (e.g., 7 for the example classification: from 0 = dry to 6 = $v \geq \text{class_5}$ m/s)	Mandatory

Metrics calculation

Name of the variable	Signification	Status
target_habitat	Target habitat type for which the metrics will be calculated. Metrics are only calculated for the meshes that experience this habitat type at least once during the habitat time-series Unique integer from the habitat classification definition	Mandatory
metrics_to_compute	Metrics selection: "true" to compute the metric, "false" otherwise	Mandatory
shift_all_daily	Daily shifts between all types of habitats	Mandatory
shift_targ_daily	Daily shifts between the suitable_hab_shifts habitat types	Mandatory
shift_dry_daily	Daily shifts to dry conditions	Mandatory
dry_max	Maximum consecutive time steps of dry conditions (in number of timesteps)	Mandatory
desiccation_risk	Associated desiccation risk value (see section 4.4 Metrics calculation)	Mandatory
drift_percentile	Drift percentile risk (see section 4.4 Metrics calculation)	Mandatory
drift_max	Associated maximum drift value (see section 4.4 Metrics calculation)	Mandatory
drift_durations	Number of timesteps spent in each drift category	Mandatory

suitable_hab_shifts	List of habitat types considered "suitable" in between which shifts will be calculated --> shift_targ_daily	Mandatory
start_at_first_occurrence	"true" to start the desiccation metric calculation after the first occurrence of the habitat type in the habitat time-series. "false" to start the desiccation metric calculation at the beginning of the time-series	Mandatory
desiccation_thresholds	Example of desiccation thresholds based on consecutive durations spent outside of water (in hours)	Mandatory
up_ramp	"true": drift_thresholds_with_ramp selected for drift calculation. "false", drift_thresholds selected (no consideration of ramp rate)	Mandatory
drift_thresholds_with_ramp	Example of drift thresholds based on current velocity thresholds in m/s, if up_ramp is true (see section 4.4 Metrics calculation)	Based on the value of up_ramp
drift_thresholds	Example of drift thresholds based on current velocity thresholds in m/s, without consideration of ramp rate if up_ramp is false (see section 4.4 Metrics calculation)	



It is the user's responsibility to determine the desiccation and drift threshold values that are adapted to the macroinvertebrate taxa and the habitat type of interest in the study. Moreover, the user needs to decide if an up-ramping threshold makes sense for the drift risk calculation.

Optional: advanced parameters for a focus on a specific zone

Name of the variable	Signification	Status
focus_on_zone	"true" if focus set on a specific zone (e.g., gravel banks for eggs desiccation), "false" otherwise (by default)	Mandatory
zone_interest_filename	name of the focus zone shapefile located in the data/input/Focus_zone folder	Optional
id_polygon_zone_interest	shapefile's id column name of the polygons of interest (e.g., individual gravel bar id) /!\ This column needs to be named differently than the shp_id_colname id in the input shapefile. Rename the column if needed	Optional
zone_ponte_field	name to give to this focus zone in output files (value of 1 if mesh part of the focus zone, 0 if not)	Optional

Optional: files paths for plots and clustering (further analysis)

Name of the variable	Signification	Status
base_output_path	output folder (located in data > output folder) where data to analyze is located in	Optional

final_csv_path	Relative path to the output csv file containing the metrics values	Optional
final_shp_path	Relative path to the output shapefile containing the metrics values	Optional
static_habitat_csv_path	Relative path to the csv file containing the habitat classification of each mesh	Optional

Creating the local.yaml and tailoring an analysis

Once the example case study runs successfully, you can then create a *local.yaml* file (by copy/pasting the *config.yaml*, see video) and try modifying one parameter at a time.

Simple modifications:

- Change the target habitat
- Modify one habitat threshold
- Replace the flow time-series and adapt the timestep accordingly
- Activate or deactivate one metric

More advanced modifications:

- Change desiccation thresholds
- Change drift thresholds
- Run *plots.py* or *clustering.py*

Modifications to change the focus of the analysis: focusing on a zone

- Set *focus_on_zone* to true
- Add a focus zone shapefile
- Adapt the habitat classification method to the scope of the analysis

As previously mentioned, make sure to have saved your changes (file *local.yaml* → save or simple Ctrl+S), and have saved your previous *Toolbox_run_output* outputs in a separate folder before running the *main.py*.

4. Full workflow explanation

This chapter explains what happens inside *main.py*, from input data to final metrics.

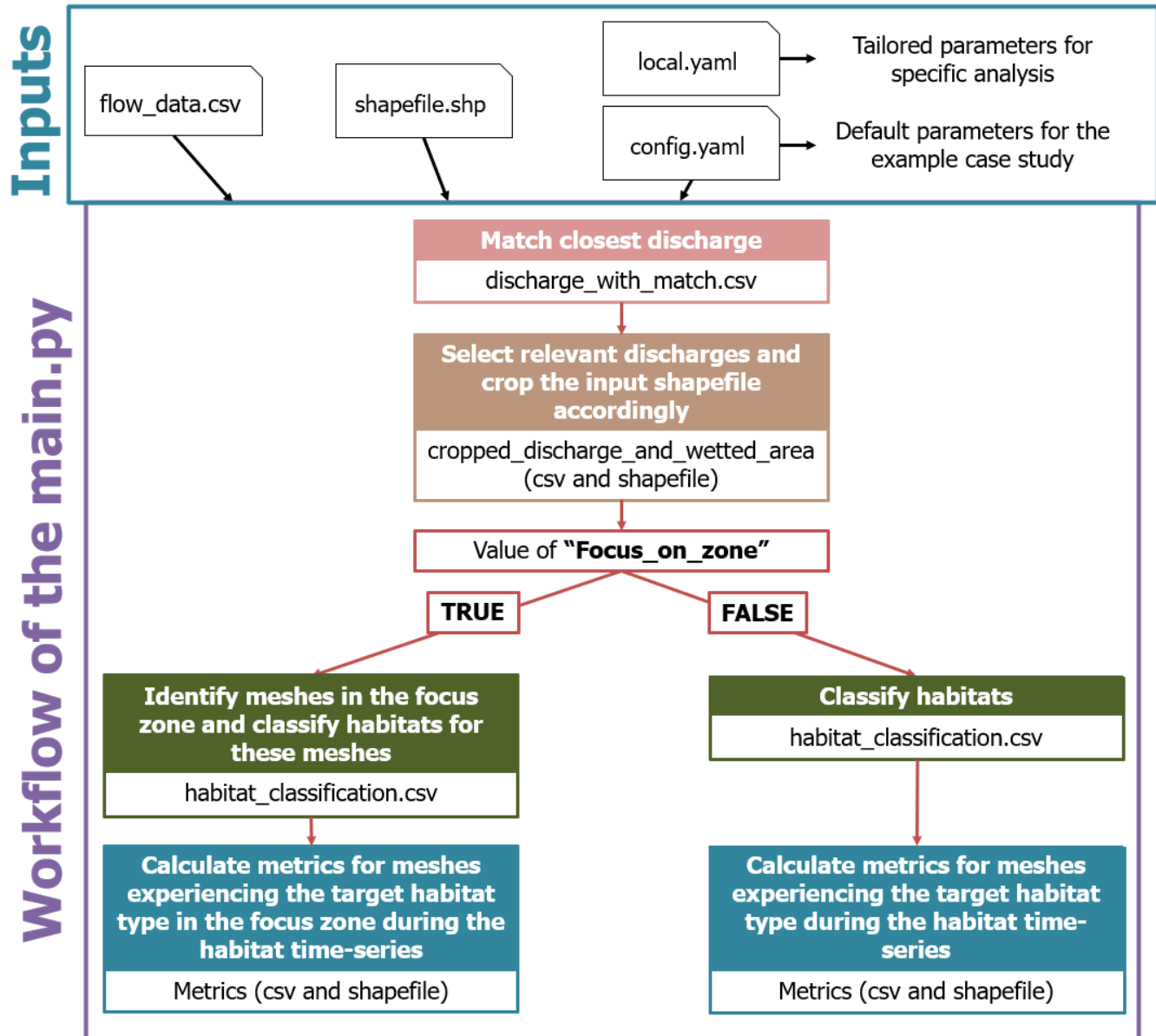


Figure 7 Full workflow of the *main.py* file

4.1 Match flow time-series to the closest simulated discharges

The toolbox associates the discharges from the input flow time-series to the closest one for which current velocity and depth data is known (data from the input shapefile, see section 3.1 Flow time-series input data). The column "Corresponding_known_discharge" will then be added as seen on Figure 8.

Datetime	Discharge	Corresponding_known_discharge
1/6/2025 0:00	1.597203229	2
1/6/2025 0:10	2.260468285	2
1/6/2025 0:20	11.27516493	10.6
1/6/2025 0:30	13.44403064	13.4
1/6/2025 0:40	11.49950344	12
1/6/2025 0:50	9.056749663	9

Figure 8 Example of processed flow time-series where each discharge is associated to the closest known one

4.2 Select relevant discharges and crop the input shapefile

From these “Corresponding_known_discharge”, the toolbox gets the list of the discharges to keep for the scope of the study. To save computational time, input files get cropped accordingly to keep:

- Only data corresponding to discharge in the scope of the study
- Only meshes that are wet under the maximum considered discharge

4.3 Classify habitats

It is the user’s responsibility to define how to classify habitat types to match the scope of the analysis. Figure 9 presents the water depth and current velocity-based classification used in the example case. Similar approaches can be used with a varying number of habitat types and different threshold values, **all to be defined and adapted in the configuration file (*local.yaml*) and the *habitat_classification.py* file.**

Color	Current Velocity Class
	Dry
	Stagnant to slow current $v_{40} < 5$ cm/s
	Slow current $5 \leq v_{40} < 25$ cm/s
	Slow to fast current $25 \leq v_{40} < 75$ cm/s
	Fast to very fast current $75 \leq v_{40} < 150$ cm/s
	Very fast current $150 \leq v_{40} < 250$ cm/s
	Violent current $v_{40} \geq 250$ cm/s

Figure 9 Example of macroinvertebrate habitat classification at community-level (adapted from Tonolla 2023)

Another example of habitat classification focusing on a specific zone (with the boolean *focus_on_zone* set to “true”) is provided in the function *attribute_habitat_types_zone_only* in the *habitat_classification.py* script. This example targets egg deposition on gravel banks. Habitat type -1 indicates that the mesh element lies outside the zone of interest, while the remaining habitat types are illustrated in Figure 10.

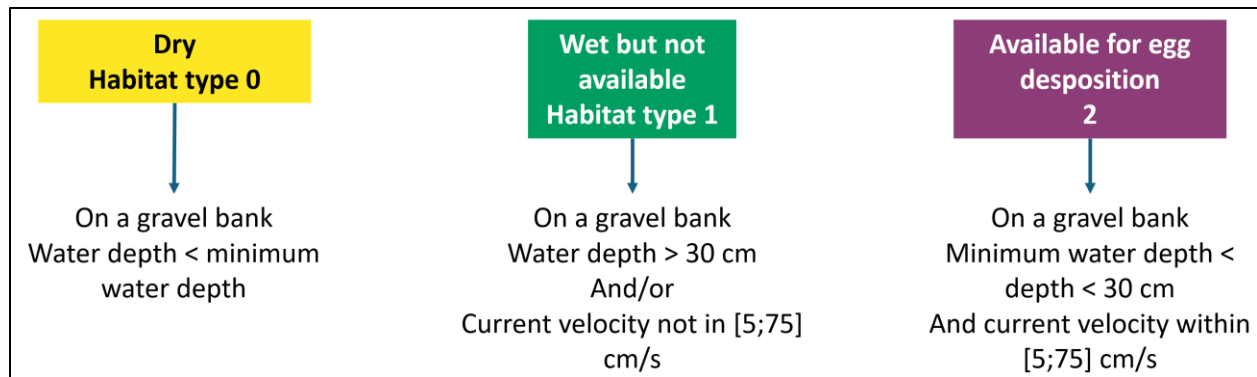


Figure 10 Example of habitat classification focusing on egg desiccation on gravel banks



When adapting the code in the repository (e.g., *the habitat_classification.py*), make sure to **save your changes before running the code** for your changed to be considered (simple Ctrl+S in the file).

As previously mentioned, also make sure to save your changed files to a safe location before pulling the latest version of the Git repository.

4.4 Metrics calculation

Scope of the metrics calculation

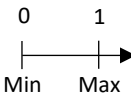
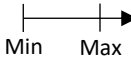
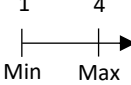
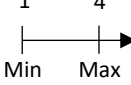
The simulated domain includes all patches that are wet under the maximum discharge of the flow time-series. The habitat probabilities are calculated for all these wetted patches.

The metrics are calculated only for patches that experience the habitat type in focus during the habitat time-series. The user can decide which metric(s) to calculate in the configuration file.

Description of the metrics

The macroinvertebrate patch-scale habitat metrics on Table 5 Overview of the four patch-scale metrics used to study macroinvertebrate habitat dynamics, including definitions and calculation are derived from the habitat time-series to quantify complementary ecological facets of habitat dynamics experienced by macroinvertebrates. Habitat probability (M1) and shifts (M2) describe the availability and temporal persistence of suitable conditions and are therefore generally related to habitat quality, whereas drift (M3) and desiccation (M4) metrics represent exposure to harmful hydraulic events with associated ecological risk.

Table 5 Overview of the four patch-scale metrics used to study macroinvertebrate habitat dynamics, including definitions and calculation

Metric	How ?	What ?	Why ?
1) Habitat probability	Likelihood of different habitat conditions within a patch through time (Bätz et al. 2025). The value is continuous. Value = $\frac{\sum \text{duration of occurrence of a specific habitat type over the time-series}}{\text{total time of the flow time-series}}$		- Time-integrated habitat availability - Helps to spot dominant habitat types
2) Habitat shifts	Daily number of times (i.e., frequency) that habitat conditions change to another suitable habitat type in a patch (Bätz et al. 2025). The value is continuous. Value = $\frac{\sum \text{number of relevant shifts during the considered time-series}}{\text{number of days of the flow time-series}}$	Daily time intervals 	- Highlights habitat stability and persistence
3) Drift risk	Risk of macroinvertebrate drift based on current velocity conditions and water level rise experienced within each patch over time. The value is discrete. Up-ramping can be included in the analysis or not Value = 90th percentile risk value reached over the habitat time-series		- Identifies exposure to critical passive drift
4) Desiccation risk	Risk of eggs desiccation linked to hydropeaking-induced water level variations. The value is discrete, and calculated from start or after a first occurrence Value = maximum risk value reached over the habitat time-series <i>Error! Reference source not found.</i>		- Identifies exposure to critical desiccation duration

Drift and desiccation risks are based on threshold values, that depend not only on the considered taxa, but also on the available data and the habitat type in focus.

Table 6 and Table 7 are examples of threshold values used in example case.

Table 6 focuses on the drift risk for two habitat types: Slow (habitat type 2) and Slow to fast (habitat type 3) current velocity. “Slow” habitat type considers the notion of up-ramping rate (up_ramp set to “true” in the configuration file, and a value entered in drift_ramp_threshold). The analysis of the habitat type “Slow to fast” has no up_ramp. The parameters set in the configuration file allow to pick for the right drift calculation function in the metric calculation scripts.

Table 7 presents the example of egg desiccation risk threshold values implemented in the toolbox.

Table 6 Macroinvertebrate drift risk thresholds and corresponding metric values for two considered habitat types

Metric Value	Drift risk class	Habitat type		
		Slow ^a		Slow to fast ^b
		Current velocity threshold	Water level rise	Current velocity threshold
1	Low	$v_{40} \leq 0.4$ m/s	Not relevant	$v_{40} \leq 1.2$ m/s
2	Moderate	$0.4 < v_{40} < 0.7$ m/s	Not relevant	$1.2 < v_{40} < 2.1$ m/s
		$0.7 \leq v_{40} \leq 1$ m/s	< 1cm/min	
3	High	$0.7 \leq v_{40} \leq 1$ m/s	≥ 1 cm/min	$2.1 \leq v_{40} \leq 3$ m/s
4	Very high	$v_{40} > 1$ m/s	No data	$v_{40} > 3$ m/s

^a: thresholds taken from the flume experiments in Tonolla et al. (2019) and incorporating findings from Friese et al. (2025).

^b: thresholds mathematically extrapolated from the “slow” current habitat, by proportionally adjusting the upper limit of the typical current velocity range. No water level rise rate included to avoid introducing additional assumptions.

Table 7 Macroinvertebrate desiccation risk thresholds and metric values based on Kennedy et al. (2016) and Jaulin (2025)

Metric value	Desiccation risk class	Max duration dry
1	Low	≤ 1 hour
2	Moderate	1 - 5 hours
3	High	5 - 9 hours
4	Very high	> 9 hours

The *start_at_first_occurrence* boolean can change the values of the desiccation risk: if set to “true”, desiccation risk will only be calculated after the first occurrence of the target habitat type. If set to false, the calculation starts at the beginning of the flow time-series. For example, for a study focusing on egg desiccation on gravel banks, setting the *start_at_first_occurrence* to true will make the desiccation risk calculation start after the first occurrence of the target habitat type that is “suitable for egg deposition”.

Advanced metrics when focusing on a certain zone

In the advanced case when *focus_on_zone* is true, additional metrics such as median and quartile values are being calculated for the habitat type in focus and the dry conditions to have a deeper understanding of the habitat probability distributions (see the Table 11 in section 5.1 Outputs from the main file execution, and the related part of section 5.2 Outputs from the plot execution).

4.5 Estimation of the computational time

The computational time can drastically vary depending on the spatial and temporal coverage of the analysis (see estimations on Figure 8).

Table 8 Estimation of computational times based on the number of meshes and of timesteps included in the analysis

Number of meshes (total)	Number of meshes for metrics calculation	Number of timesteps	Estimated computational time
365 000	27 350	1008	~ 50 minutes
664 000	76 060	4393	~ 4 hours

4.6 Additional functions

Table 9 Association between the code to run and the desired outputs

Output to get	Python file to run
Metrics results	<i>main.py</i>
Plots	<i>plots.py</i>
Clustering	<i>clustering.py</i>

Before running the additional scripts, paths to the files to study (*base_output_path*, *final_csv_path*, *final_shp_path*, *static_habitat_csv_path*) need to be entered in the configuration file (see section 3.2), which then needs to **be saved** before running the scripts.

The metrics in the scope of the plot and the clustering functions are taken from the configuration file. This means you must ensure the metrics selected match the ones that are in the files to analyze. E.g., if the boolean *desiccation_risk* is set to true in the configuration file, it also needs to be in the file to analyze because the functions will be searching for it.

Plots

Outputs location: in data > outputs > base_output_path > Plots

Plots focus on two geographical extents:

- The entire simulated domain, for which habitat probabilities and most probable habitat types are assessed. This helps understanding the global picture and understand why certain meshes do not experience the focus habitat type during the habitat time-series.
- The domain for which metrics are calculated, including only meshes that experience the focus habitat type at least once during the habitat time-series.

If more targeted plots are needed, the user is free to modify the functions or add new ones (!\ as previously mentioned, pulling the latest version of the repository from GitHub will overwrite everything so make sure to save your changes in a safe folder).

Examples of outputs are in section 5.2 Outputs from the plot execution.

Clustering

Outputs location: in data > outputs > base_output_path > Clustering

This script conducts a k-means clustering to synthesize the combined information contained in the four metrics. The purpose is to group patches with similar multimeric profiles, thereby summarizing complex combinations of macroinvertebrate habitat quality and risk into a reduced number of ecologically interpretable groups. Because each metric represents a distinct dimension of macroinvertebrate habitat dynamics, clustering enables the identification of patterns that are not apparent when examining each metric separately. The optimal number of clusters is determined using the silhouette score, with maximum numbers of clusters set to 9 (can be adapted in the code if needed).

The functions provide details on the cluster statistics, on the Principal Component Analysis (PCA) loadings, and adds the cluster number of each mesh in the csv and shapefile metrics documents. More information on the output files is available on section 5.3).

5. Outputs interpretation

Outputs location: data > output > base_output_path folder. Relevant subfolders separate the outputs.

5.1 Outputs from the main file execution

The *main.py* file execution provides the outputs presented in Table 10.

Table 10 Outputs of the main.py file

Output name	Output location	Explanation
discharge_with_match.csv	Discharge_with_match	Displays each discharge of the flow time-series with its corresponding known simulated discharge.
cropped_discharge_and_wetted_area (csv and shapefile)	Cropped_shapefiles	Input shapefile cropped to the considered discharges and the associated spatial area (meshes that are wet under the maximum discharge considered). This allows to reduce the zone of interest before processing the metrics to reduce the computational time.
habitat_classification.csv	Habitat_classification	Static picture of habitat classification for each mesh and each considered discharge before time integration. This file is used for the plots to represent habitat availabilities at each discharge.
metrics (csv and shapefile)	Metric_files	Files having the final metrics results for each mesh experiencing the habitat target during the habitat time-series.



The shapefile column names are limited to 10 characters. Table 11 provides more details on shapefile column names that had to be shrunk.

Table 11 Explanation of the different columns that can be found in the metric outputs (csv and shapefile)

Id	Id of the mesh (identical as input shapefile)
x	X coordinate of the mesh (identical as input shapefile)
y	Y coordinate of the mesh (identical as input shapefile)
Area	Area of the mesh (identical as input shapefile)
d_discharge	Depth value for each discharge in scope of the current analysis (identical as input shapefile)
v_discharge	Current velocity value for each discharge in scope of the current analysis (identical as input shapefile)

focus_zone	Only if there is a focus on a certain zone (focus_on_zone is true). 0 if not in the interest zone shapefile, 1 if in the focus zone shapefile
id_focus	Only if there is a focus on a certain zone (focus_on_zone is true). Indicates the id of the polygon of the interest zone shapefile (e.g., the individual id of the gravel bank)
Habitat probability	
dur_h	Duration of each habitat (in number of timesteps) during the habitat time-series The amount of these columns depends on the habitat classification: - From -1 to 3 for the egg desiccation habitat classification example - From 0 to 6 for the example habitat classification based on water depth and current velocity
prob_h	The associated habitat probabilities = duration habitat type/total duration (in number of timesteps)
mostProb	Most probable habitat type per mesh over the flow time-series. It is the habitat type having the highest habitat probability for each mesh
h_first	Timestep number of first occurrence of the habitat type considered. e.g., h2_first: timestep number of the first occurrence of the habitat type 2 for this mesh over the habitat time-series
Statistics on the target habitat type – only for advanced studies focusing on a certain zone	
h_nb_seq	Number of separated sequences of target habitat type conditions in the mesh over the habitat time-series
h_dur_max	Maximum consecutive duration (in hours) of target habitat type conditions in the mesh over the habitat time-series
h_dur_med	Median of the consecutive duration (in hours) of target habitat type conditions in the mesh over the habitat time-series
h_dur_q1	1 st quartile of the distribution of the consecutive duration of target habitat type conditions in the mesh over the habitat time-series
h_dur_q3	3 rd quartile of the distribution of the consecutive duration of target habitat type conditions in the mesh over the habitat time-series
Habitat shifts	
h_sh_all	Total daily shifts experienced by this mesh over the habitat time-series
h_sh_suit	If not focusing on a zone: total daily shifts to the suitable habitat types (entered in the configuration file) over the habitat time-series. If focusing on a zone: total daily shifts to the target habitat over the habitat time-series
h_sh_dry	Total daily shifts to dry conditions over the entire habitat time-series
Desiccation	
h_dryMax	Maximal duration of consecutive dry conditions (in number of timesteps).

	The start of the calculation depends on if <code>start_at_first_occurrence</code> is “true” or “false”: calculation will start at the beginning of the habitat time-series if “false”, and only after the first occurrence of the target habitat type if “true”
<code>h_desicR</code>	Associated desiccation risk
Desiccation additional statistics – only for advanced studies focusing on a certain zone	
<code>h_dry_seq</code>	Number of separated sequences of dry conditions in the mesh over the habitat time-series
<code>h_dry_max</code>	Maximum consecutive duration of dry habitat type conditions in the mesh over the habitat time-series
<code>h_dry_med</code>	Median of the consecutive duration of dry habitat type conditions in the mesh over the habitat time-series
<code>h_dry_q1</code>	1 st quartile of the distribution of the consecutive duration of dry habitat type conditions in the mesh over the habitat time-series
<code>h_dry_q3</code>	3 rd quartile of the distribution of the consecutive duration of target habitat type conditions in the mesh over the habitat time-series
<code>h_dryMedR</code>	Risk value associated to the median of the consecutive duration of dry habitat type conditions in the mesh over the habitat time-series
<code>h_dry_q1R</code>	Risk value associated to the 1 st quartile of the distribution of the consecutive duration of dry habitat type conditions in the mesh over the habitat time-series
<code>h_dry_q3R</code>	Risk value associated to the 3 rd quartile of the distribution of the consecutive duration of target habitat type conditions in the mesh over the habitat time-series
Drift	
<code>h_dd_driftRisk</code>	Drift duration of each drift risk category (in number of timesteps)
<code>h_driftP</code>	Associated drift risk percentile
<code>h_driftM</code>	Associated maximal drift risk value

5.2 Outputs from the plot execution

Output location: `data > output > base_output_path > Plots`

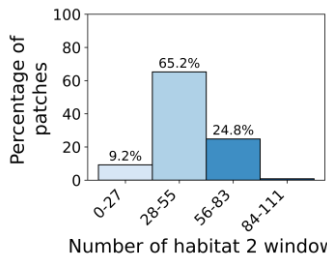
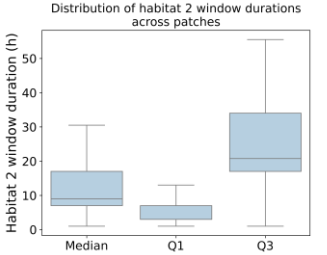
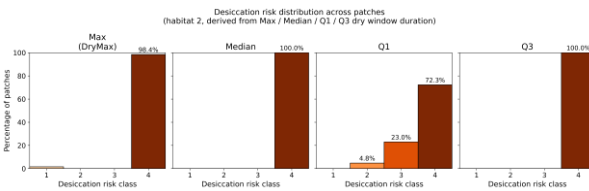
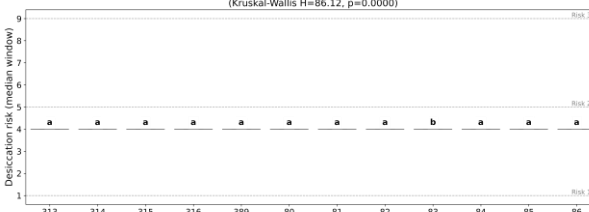
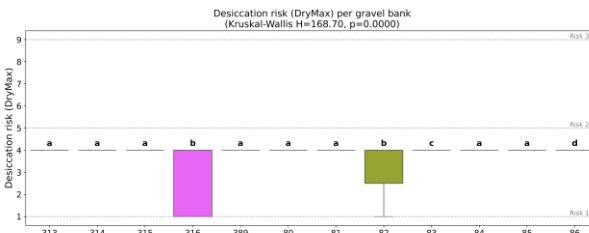
The description of the plot function is available in section 4.6 Plots. The user can always modify the function to tailor plots to specific needs, add or delete some. As previously mentioned, the updated *plot.py* needs to be saved in a separate file not to be overwritten when pulling the latest version of the GitHub.

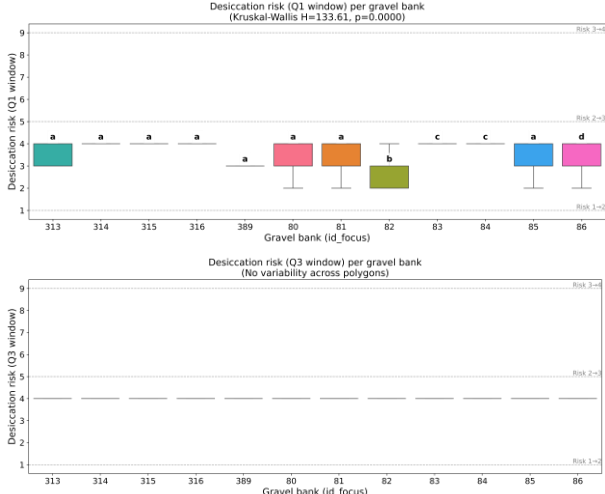
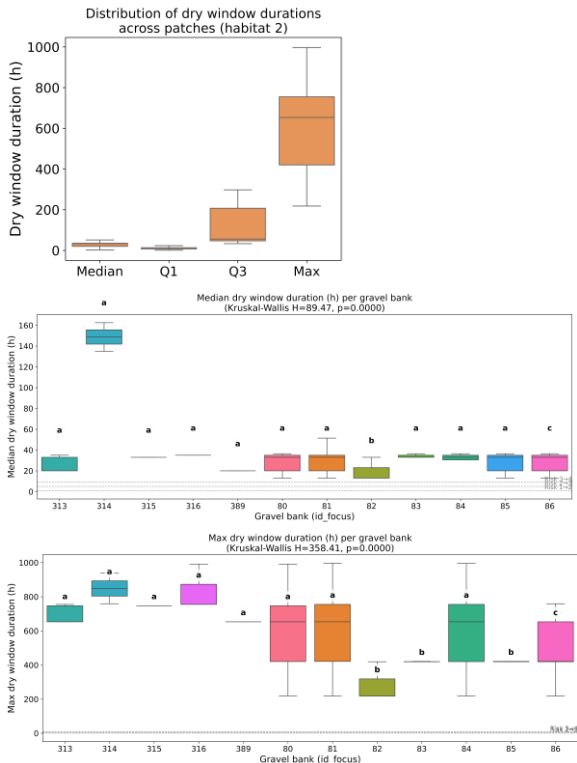
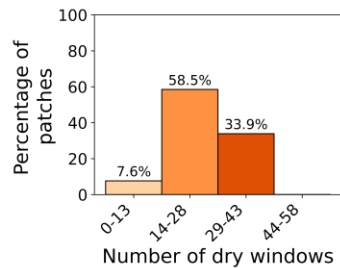
The outputs depend on the values of the parameters in the *local.yaml* configuration file. For the case when meshes have different sizes, some plots normalize by area.

Default plot outputs are presented for the example case study in the following section 5.4 Example case study: output presentation.

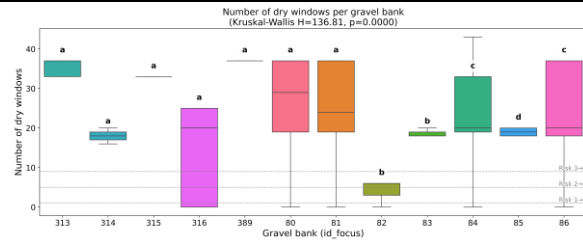
For advanced analysis such as the ones focusing on zones (Boolean *focus_on_zone* is true), additional plots can be gone as presented on the below. To compare the different gravel banks, it means that all meshes have an id corresponding to which gravel bank they belong to.

Table 12 Advanced plots done by the file plot.py when focusing on a specific zone

Target habitat	
Number of target habitat sequence windows	 Percentage of patches Number of habitat 2 window
Distribution of target habitat sequences durations	 Distribution of habitat 2 window durations across patches Habitat 2 window duration (h) Median Q1 Q3
Dry durations and desiccation risk	
Desiccation risk	
Distribution of the risk values across patches	 Desiccation risk distribution across patches (habitat 2, derived from Max / Median / Q1 / Q3 dry window duration) Percentage of patches Desiccation risk class Desiccation risk (median window) per gravel bank (Kruskal-Wallis H=86.12, p=0.0000)
Median desiccation risk per gravel bank	 Desiccation risk (median window) Gravel bank (id_focus)
Desiccation risk per gravel bank	 Desiccation risk (DryMax) per gravel bank (Kruskal-Wallis H=168.70, p=0.0000) Desiccation risk (DryMax) Gravel bank (id_focus)

Q1 of the desiccation risk per gravel bank Q3 of the desiccation risk per gravel bank	 Desiccation risk (Q1 window) per gravel bank (Kruskal-Wallis H=133.61, p=0.0000) Desiccation risk (Q3 window) per gravel bank (No variability across polygons)
Associated dry window durations Median dry window duration (in hours) per gravel bank Maximum dry window duration (in hours) per gravel bank	 Distribution of dry window durations across patches (habitat 2) Median dry window duration (h) per gravel bank (Kruskal-Wallis H=89.47, p=0.0000) Max dry window duration (h) per gravel bank (Kruskal-Wallis H=358.41, p=0.0000)
Number of dry windows	 Percentage of patches Number of dry windows

Number of dry windows per gravel bank



5.3 Outputs of the clustering execution

Output location: data > output > base_output_path > Clustering

Description of the plot function available in section Additional functions 4.6 Clustering.

The outputs depend on the values of the parameters in the *local.yaml* configuration file.

Outputs are presented for the example case study in the following section 5.4 Example case study: output presentation.

5.4 Example case study: output presentation

Metrics outputs and visual representation

A QGIS project representing the metrics outputs is available in the data > output > Example_output > QGIS_project folder. An explanatory video on the topic is also available.

Example case study: Metrics outputs and visual representation

As presented on , the most probable habitat type over the habitat time-series are represented in a layer, such as the four metrics for the default parameters from the *conifg.yaml* file.

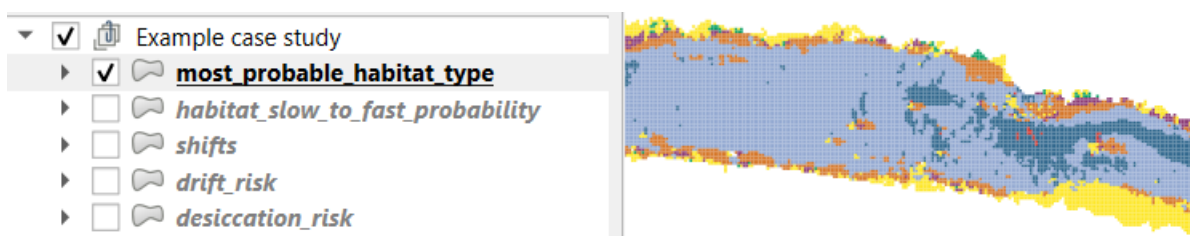


Figure 11 QGIS layers present for the example case study

Plot outputs

Example case study: Example case study plots

below presents examples of plots done by the file *plot.py* for the example case study.

Table 13 Plots done by the file *plot.py* for the example case study

Static representation of habitats availability per discharge		
	For the entire simulated domain	For the metrics domain
Per number of patches		

Per area		
Integration in time and space		
Most probable habitat type distribution (per number of patches)		
Associated probabilities		
Habitat probabilities distributions of patches experiencing the focus habitat type		
Metrics representations		
Habitat probability		
Daily shifts to targeted habitats		

Desiccation risk		<table><tr><th>Desiccation risk</th><th>Percentage of patches</th></tr><tr><td>1</td><td>52.3%</td></tr><tr><td>2</td><td>33.7%</td></tr><tr><td>3</td><td>9.6%</td></tr><tr><td>4</td><td>4.4%</td></tr></table>	Desiccation risk	Percentage of patches	1	52.3%	2	33.7%	3	9.6%	4	4.4%
Desiccation risk	Percentage of patches											
1	52.3%											
2	33.7%											
3	9.6%											
4	4.4%											
Drift risk		<table><tr><th>Drift risk</th><th>Percentage of patches</th></tr><tr><td>1</td><td>36.0%</td></tr><tr><td>2</td><td>63.0%</td></tr><tr><td>3</td><td>1.0%</td></tr><tr><td>4</td><td>0.0%</td></tr></table>	Drift risk	Percentage of patches	1	36.0%	2	63.0%	3	1.0%	4	0.0%
Drift risk	Percentage of patches											
1	36.0%											
2	63.0%											
3	1.0%											
4	0.0%											

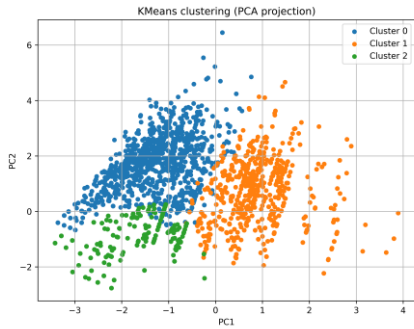
Clustering outputs

Example case study: Example case study clustering

below presents examples of plots done by the file *clustering.py* for the example case study.

Table 14 Clustering outputs done by the file clustering.py for the example case study

Output	Description																		
Hab_silhouette.png	Plot showing the evolution of the silhouette score with the number of clusters. The maximum value determines the optimal number of clusters. <table border="1"> <thead> <tr> <th>Number of clusters</th> <th>Score</th> </tr> </thead> <tbody> <tr> <td>2</td> <td>0.48</td> </tr> <tr> <td>3</td> <td>0.49</td> </tr> <tr> <td>4</td> <td>0.37</td> </tr> <tr> <td>5</td> <td>0.39</td> </tr> <tr> <td>6</td> <td>0.40</td> </tr> <tr> <td>7</td> <td>0.39</td> </tr> <tr> <td>8</td> <td>0.40</td> </tr> <tr> <td>9</td> <td>0.35</td> </tr> </tbody> </table>	Number of clusters	Score	2	0.48	3	0.49	4	0.37	5	0.39	6	0.40	7	0.39	8	0.40	9	0.35
Number of clusters	Score																		
2	0.48																		
3	0.49																		
4	0.37																		
5	0.39																		
6	0.40																		
7	0.39																		
8	0.40																		
9	0.35																		
Hab_PCA_clusters.png	Graphically represents the position of the clusters on the PC1/PC2 axis																		

	
Hab_PCA_loadings.csv	Provides the value of PC1 and PC2 loadings for each metric
metrics_with_clusters (csv and shapefile)	Same outputs as the metric results ones, having additional columns: - Cluster_hab: indicating the cluster number of each mesh - PC1_hab and PC2_hab: principal component loadings of the mesh

6. Limitation and assumptions

Input shapefile: it is assumed that the user has identified key study discharges and conducted a hydrodynamic simulation to get the required input shapefile.

Habitat classification and metrics threshold values: the toolbox only provides example threshold values that need to be tailored to the user needs and taxa of interest.

GIS analysis: the visual analysis of the output shapefile is to be done on a GIS tool by the user.

7. Trouble shooting

Here is a list of recommendations to get familiar with the toolbox and avoid troubles.

7.1 Start small, in space and in time

Start with small spatial zones and short flow time-series: this will not take long to compute, and you will still be able to validate the results with common sense before it gets too complex.

To help you crop the original shapefile to a smaller zone to get started, the QGIS procedure is explained in the section 9.2 Crop a .shp to a specific shape in QGIS.

7.2 Save your scripts and analysis files

As previously mentioned, when pulling the latest version of the GitHub repository, your current data and scripts will be overwritten. Make sure to save your files in a separate folder not to lose anything.

7.3 Follow the instructions carefully

When it is said that the column names format matters, it is because the code extracts an information out of them. For instance, if your input shapefile column names are not `depth_18_3` for the discharge $18.3 \text{ m}^3/\text{s}$ (as requested), but entered as `depth_18.3`, the code will crash, and you will have no outputs. You can simply modify your column names with your usual GIS tool such as QGIS.

Moreover, if you do not create the *local.yaml* file, the toolbox will run with the default parameters (corresponding to the example case study). Make sure to respect the necessary steps to tailor and conduct specific analysis.

8. Glossary

Patch: is a spatially fixed unit within a river reach, typically corresponding to areas on the order of 1 to 100 m² (micro- to meso-scale), with the appropriate size depending on the mobility, species, and life stage considered. While the location of a patch remains constant, the habitat conditions within it may vary over time. In this toolbox, the patches usually match the meshes of the hydrodynamic simulation for which depth and current velocity data is available.

Habitat: refers to a set of environmental conditions characterized by dominant physical and chemical properties (e.g., current velocity, water depth, substrate). These conditions can be described at different spatial scales, including the micro- to meso-scale (1–100 m²) within a river reach. Different species and life stages rely on specific habitat conditions to fulfil essential functions such as feeding, reproduction, resting, or refuge. The spatial distribution of habitats within a river reach can change over time.

Habitat shift: occurs when environmental conditions change in space and/or time, leading to a transition from one habitat type to another. In this toolbox, habitat shifts are assessed at the patch scale, capturing temporal changes in habitat conditions within a fixed spatial unit.

Flow time-series: describes the temporal evolution of river discharge (m³/s), represented at a defined timestep (e.g., minutes or hours).

Habitat time-series: represents the temporal sequence of habitat conditions within each patch, obtained by combining spatial habitat classifications with the flow time-series.

Target habitat: The habitat type that is the primary focus of the analysis. Habitats are defined according to a chosen classification scheme, where each type is assigned a numerical code for processing within the toolbox. Metrics are computed specifically for mesh elements that experience the target habitat at least once during the habitat time-series.

Suitable habitats: Habitat types that are considered favourable for the macroinvertebrate taxa under study, based on the criteria defined in the selected habitat classification scheme.

Drift percentile (drift risk): A patch-scale metric representing exposure to hydraulically induced drift. It is defined as the 90th percentile of all drift-risk values computed from the habitat time-series for a given patch. This percentile-based approach captures frequent exposure to elevated drift conditions while reducing the influence of isolated extreme events.

Desiccation risk: A patch-scale metric representing exposure to drying conditions over time. It is derived from the habitat time-series and reflects the occurrence and persistence of conditions where habitat becomes unsuitable due to insufficient water depth, indicating ecological stress for macroinvertebrates.

9. QGIS tips and tricks

9.1 Analysing a shapefile field on QGIS

To focus on a specific shapefile field, the Figure 12 Example of how to process the output shapefiles on QGIS is an example of how to do it on QGIS.

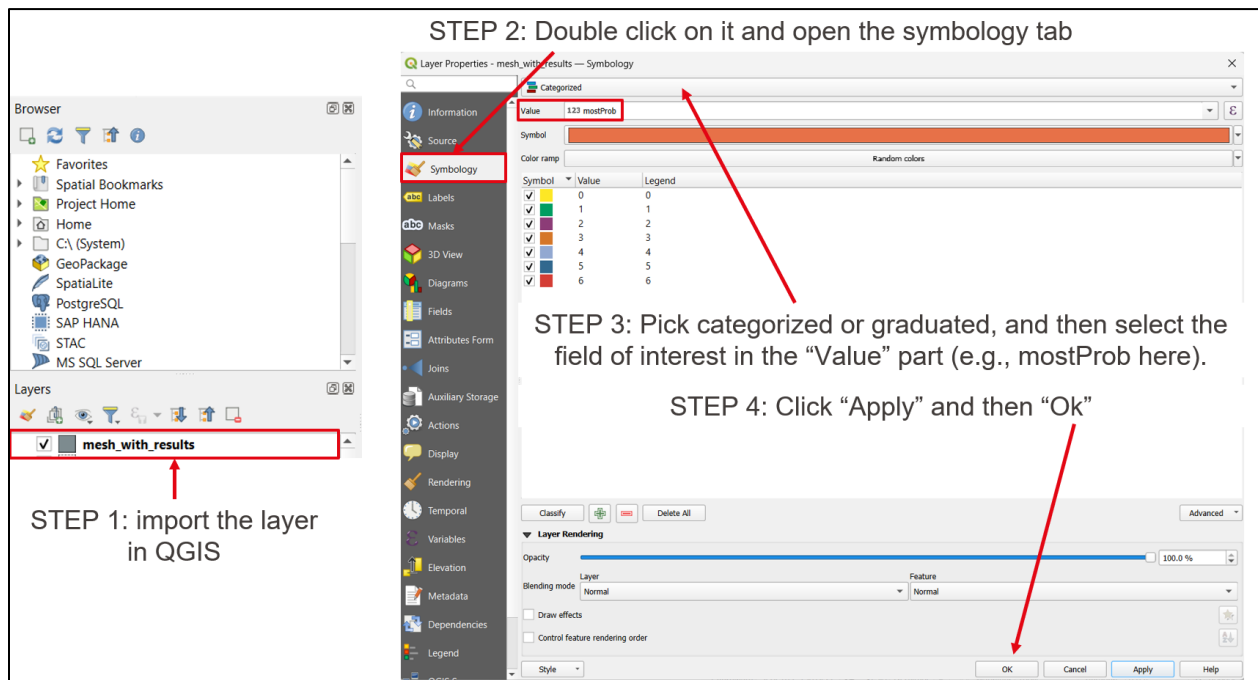


Figure 12 Example of how to process the output shapefiles on QGIS

9.2 Crop a .shp to a specific shape in QGIS

This gives a reduced .shp file containing the data for the area of interest.

Step 1. Prepare your layers

- You need two layers:
 - The .shp layer to crop (the one you want to limit to a certain area).
 - The mask layer (the polygon defining the area you want to crop to).

Step 2. Use the "Clip" tool

- Open the Processing Toolbox (from the menu: Processing → Toolbox).
 - Search for "Clip".
 - Choose Vector overlay → Clip.
 - In the dialog:
 - Input layer: the shapefile you want to crop.
 - Overlay layer: the polygon mask.
 - Save to file: choose a path for the new cropped shapefile.
 - Click Run.
- The resulting shapefile will only contain features inside your mask polygon.

Step 3. About features with missing data

- If parts of the shapefile inside your mask have null or missing attribute values, the clip will still work.
- QGIS doesn't require all attributes to be filled; missing data won't stop the crop.
- However, if you need to do further analysis (e.g., statistics or joins), you may need to handle nulls.

Step 4. Optional: use "Extract by location"

- If you only want features fully within your mask polygon, you can use Vector → Geoprocessing Tools → Extract by location, choosing "within" instead of "intersect".
- This avoids partially overlapping features if you only want complete ones.

9.3 Joining layers on QGIS

To be done when two input shapefiles need to be joined into a single one (e.g., one is a point .shp having depth and velocities values for each mesh, the other one is a polygon .shp just containing the shape of the meshes). Both have the same center of the polygons, to be able to join the attributes by location.

The desired output is one polygon .shp file having data for each of the mesh.

Necessity: having a common id between the two input shapefiles to join based on it.

How to do: Perform the "create spatial index" for both layers: layer > source > geometry > create spatial index.

Then in the processing toolbox, process the function: **"join attributes by location"**.

10. Appendix

10.1 Deep description of files and functions

Main.py

The **main.py** file collects functions from other python files and variables entered by the user in the configuration file and executes the code.

Description of the workflow (in chronological order)	Name of the function	Where the function is located	Name of the output
Getting the known discharge values from the column names of the input polygon shapefile	get_discharge_values	<i>support_functions.py</i>	discharge_values
Match the discharge of the flow time-series to the closest known one in discharge_values	match_closest_discharge	<i>flow_time_series.py</i>	discharge_with_match.csv
Select relevant discharges for the analysis: the list of corresponding known discharges	get_study_discharges	<i>flow_time_series.py</i>	relevant_discharges
Crop the shapefile: to the corresponding relevant discharges; and keeping only the wetted mask (meshed wet for the max considered discharge)	prepare_wetted_shapefile_for_relevant_discharges	<i>support_functions.py</i>	cropped_discharge_and_wetted_area.shp
Export the cropped shapefile to a csv	prepare_csv	<i>habitat_classification.py</i>	cropped_discharge_and_wetted_area.csv
If focus_on_zone is true			
Attribute habitat only for meshes in the focus zone (otherwise -1 by default). This habitat classification function needs to be adapted by the user	attribute_habitat_types_zone_only	<i>habitat_attribution.py</i>	habitat_classification.csv
Metrics calculation	process_mesh_data_focus_on_zone	<i>metrics_calculation_focus_on_zone.py</i>	metrics.csv

Save the metrics to a shapefile	join_mesh_with_CSV_data	support_functions.py	metrics.shp
If focus_on_zone is false			
Attribute habitats based on the example habitat classification	attribute_habitat_current_based	habitat_classification.py	habitat_classification.csv
Metrics calculation	process_mesh_data	metrics_calculation.py	metrics.csv
Save the metrics to a shapefile	join_mesh_with_CSV_data	support_functions.py	metrics.shp

Plot.py

The **plot.py** contains the function for the different plots. The user can always add and/or modify these functions (more information in section 4.6 Additional functions and section 5.2).

Description of the workflow (in chronological order)	Name of the function	Name of the output
ALL MESHES IN SIMULATED DOMAIN Plot most probable habitat type and related intensity	<i>plot_most_prob_intensity</i>	hist_most_prob_and_associated_intensity_simulated_zone.png
TARGET HABITAT Plot most probable habitat type and related intensity	<i>plot_most_prob_intensity_target_hab</i>	hist_most_prob_and_associated_intensity_target_habitat.png
ALL MESHES IN SIMULATED DOMAIN Percentage of most probable habitat types	<i>plot_most_prob_percentages</i>	most_prob_percentages_simulated_zone.png
TARGET HABITAT Percentage of most probable habitat types	<i>plot_most_prob_percentages_target_hab</i>	most_prob_percentages_target_hab.png
ALL MESHES IN SIMULATED DOMAIN Horizontal stacked bar of dominant habitat types	<i>plot_most_prob_horizontal</i>	most_prob_horizontal_simulated_zone.png
TARGET HABITAT Horizontal stacked bar of dominant habitat types	<i>plot_most_prob_horizontal_target_hab</i>	most_prob_horizontal_target_hab.png
ALL MESHES IN SIMULATED DOMAIN IN PERCENTAGE OF MESHES - Habitat availability per discharge (stacked %)	<i>plot_habitat_availability_per_discharge</i>	habitat_availability_per_discharge_simulated_domain.png

TARGET HABITAT IN PERCENTAGE OF MESHES - Habitat availability per discharge (stacked %)	<i>plot_habitat_availability_per_discharge_target_hab</i>	habitat_availability_per_discharge_target_hab.png
FUNCTION: ALL MESHES IN SIMULATED DOMAIN IN AREA – Habitat availability per discharge (stacked %)	<i>plot_habitat_availability_per_discharge_area</i>	habitat_availability_per_discharge_simulated_domain_area.png
TARGET HABITAT IN AREA – Habitat availability per discharge (stacked %)	<i>plot_habitat_availability_per_discharge_area_target_hab</i>	habitat_availability_per_discharge_simulated_domain_target_hab_area.png
TARGET HABITAT Habitat probability for each habitat type (Among all meshes that ever experience the target habitat, what is the distribution of time spent in each habitat?)	<i>plot_probability_distribution_target_hab</i>	habitat_probabilities_target_hab.png
TARGET HABITAT Plot percentage histograms for selected metrics of a target habitat	<i>plot_histograms_target_habitat</i>	histogram_prob_percentage.png histogram_shifts_percentages.png histogram_desicc_percentages.png histogram_drift_percentages.png boxplot_dry_window_duration.png boxplot_targ_window_duration.png
TARGET HABITAT Compare desiccation metrics across gravel banks (id_focus) using Kruskal-Wallis and Dunn post- hoc tests, with boxplots per polygon.	<i>plot_desiccation_by_polygon</i>	desicc_max_h_by_polygon.png desicc_median_h_by_polygon.png desicc_risk_by_polygon.png desicc_risk_median_by_polygon.png desicc_risk_q1_by_polygon.png desicc_risk_q3_by_polygon.png

		desicc_window_count_by_polygon.png desiccation_summary_statistics.csv
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Clustering.py

The ***clustering.py*** contains the function regarding the clustering presented in section 4.6 Additional functions and section 5.3 . The user can always add and/or modify this file.

Description of the workflow (in chronological order)	Name of the function	Name of the output
Performing the clustering: Finding the best number of clusters, PCA analysis, cluster statistics	<i>perform_clustering_target_habitat</i>	metrics_with_clusters.csv PCA_clusters.png PCA_loadings.csv silhouette.png
Join a GeoDataFrame (mesh geometry) with a results DataFrame and save to a new shapefile	<i>join_mesh_with_CSV_data</i>	metrics_with_clusters.shp

11. References

- Bätz N, Judes C, Vanzo D, et al (2025) Patch-scale habitat dynamics: three metrics to assess ecological impacts of frequent hydropeaking. *Journal of Ecohydraulics* 10:79–106.
<https://doi.org/10.1080/24705357.2024.2426790>
- Friese N, Mathers KL, Weber C, et al (2025) Habitat-Specific Response of Macroinvertebrate Drift to Flow Pulses: Implications for Hydropeaking Management. *Freshwater Biology* 70:e70103. 774
<https://doi.org/10.1111/fwb.70103>
- Jaulin E (2025) Looking at the reproduction of stream insects to improve our understanding of the 796 ecological effects of artificial flow variations. Université Claude Bernard Lyon 1
- Kennedy TA, Muehlbauer JD, Yackulic CB, et al (2016) Flow Management for Hydropower Extirpates Aquatic Insects, Undermining River Food Webs. *BioScience* 66:561–575. 802
<https://doi.org/10.1093/biosci/biw059>
- Lecrivain A, De Cesare G, Weber C, & Bätz N (in Review). Macroinvertebrate habitat dynamics under frequent hydropower-induced discharge fluctuations: patch-scale metrics to quantify effects of hydropeaking and morphological complexity.
- Tonolla D, Schülting L, Dossi F, et al (2019) Assessment of hydropeaking impacts on the drift of aquatic macroinvertebrates – Flume and field experiments. Final Report
- Tonolla D (Ed.) (2023) Élaboration et évaluation de mesures d’assainissement des éclusées – Nouvelles connaissances issues de la recherche et de la pratique. Rapport sur mandat de l’OFEV.